

Question:

Calculate the quantity of heat required to (a) convert a liter of H₂O at 30°C to a liter of water at 60°C, and (b) heat a 1 kg block of aluminum from 30° to 60°C. Assume the specific heat of water and aluminum is, re-spectively, 1 cal/g°C and .215 cal/g°C.

Answer Choices:

- A. 28 Kcal, 6.3 Kcal
- B. 40 Kcal, 5 Kcal
- C. 30 Kcal, 8.58 Kcal
- D. 25 Kcal, 8 Kcal
- E. 29 Kcal, 7.77 Kcal
- F. 34 Kcal, 6.9 Kcal
- G. 30 Kcal, 6.45 Kcal
- H. 32 Kcal, 7 Kcal
- I. 35 Kcal, 7.5 Kcal
- J. 31 Kcal, 6.15 Kcal

Reasoning + Answer:

<think>

To solve this problem , we need to calculate the heat required for two different processes :

- 1 . Converting a liter of water at30Â°C to a liter of water at60Â°C .
- 2 . Heating a1 kg block of aluminum from30Â°C to60Â°C .

Part (a): Heat Required to Convert a Liter of HâHâH O from 30Â°C to 60Â°C

- 1 . **Volume of Water** : 1 liter = 1000 grams .
- 2 . **Specific Heat of Water** : 1 cal/gÂ°C .
- 3 . **Temperature Change** : 60Â°C - 30Â°C = 30Â°C .

The heat required (Q) for the conversion can be calculated using the formula :

$$[Q = m \cdot c \cdot \Delta T]$$
$$[Q = 1000 \, , \, \text{g} \cdot 1 \, , \, \text{cal/g}^\circ\text{C} \cdot 30 \, , \, \text{Â}^\circ\text{C} = 30000 \, , \, \text{cal}]$$

Convert calories to kilocalories :

$$[30000 \, , \, \text{cal} \cdot \frac{1 \, , \, \text{kcal}}{1000 \, , \, \text{cal}} = 30 \, , \, \text{kcal}]$$

Part (b): Heat Required to Heat a 1 kg Block of Aluminum from 30Â°C to 60Â°C

- 1 . **Mass of Aluminum** : 1 kg = 1000 grams .
- 2 . **Specific Heat of Aluminum** : 0.215 cal/gÂ°C .
- 3 . **Temperature Change** : 60Â°C - 30Â°C = 30Â°C .

Using the same formula :

$$[Q = m \cdot c \cdot \Delta T]$$
$$[Q = 1000 \, , \, \text{g} \cdot 0.215 \, , \, \text{cal/g}^\circ\text{C} \cdot 30 \, , \, \text{Â}^\circ\text{C} = 6450 \, , \, \text{cal}]$$

Convert calories to kilocalories :

$$[6450 \, , \, \text{cal} \cdot \frac{1 \, , \, \text{kcal}}{1000 \, , \, \text{cal}} = 6.45 \, , \, \text{kcal}]$$

Final Answer :

The heat required for (a) is 30 Kcal and for (b) is 6.45 Kcal , which corresponds to option G .

<answer> G </answer>